Stopping time and force

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain how a longer stop can reduce average force.

SUCCESS CRITERIA - I can...

1. Keep momentum change fixed in a fair comparison.
2. Compare force using impulse divided by time.
3. State why cushioning does not remove momentum change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “impulse” and “interval” to explain your model or result. Sentence starters are optional.

impulse: net force acting over time; change in momentum

interval: the time or range between two stated values

average: one value representing an interval or a set of values

01 Bring back an earlier idea

A 3 kg cart moving right at 5 m/s stops in 2 s. Take right as positive.
Find signed impulse and average force.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $|F_{avg}| = |\Delta p|/\Delta t$.
For the same momentum change, more stopping time lowers average force magnitude. Bars mean magnitude.

Worked reasoning: A stopping impulse has magnitude 8 N·s. Over 0.2 s, average force is 40 N; over 0.8 s, it is 10 N. Longer stopping time lowers the average force for the same momentum change.

Your turn: Draw both force-time rectangles with equal areas.

Sentence starter: The ... in my model represents ... . Its label shows ... .

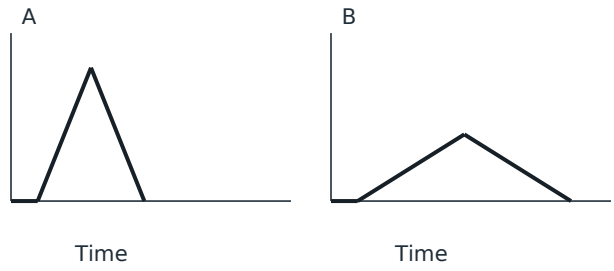
Marzano check - circle: ① ② ③

03 Represent and interpret

The two ideal force pulses have equal area.
Identify the longer-duration, lower-average-force stop.

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice ... . This supports ... because ... .



Marzano check - circle: ① ② ③

04 Solve and justify

A .060 kg ball at 20 m/s comes to rest.
Compare average stopping force for .010 s and .050 s.
State the impulse in both.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A student says a cushion makes momentum disappear. **Explain** where the momentum change comes from and what cushioning changes.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A downward-velocity fit goes from 0.4 m/s at 0.02 s to 3.34 m/s at 0.32 s.
Find g and explain why fitting an interval can be better than one noisy point.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

A 2 kg object slows from 6 to 2 m/s in .50 s.
Find impulse and average net force. If time becomes 1.0 s for the same velocity change, find the new average force.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 0.2 kg ball at 15 m/s stops in 0.03 s on one surface and 0.12 s on another.
Find average forces and design a triangular force-time model for the longer stop.

Sentence starter: My design shows ... . It meets the requirement by ... .

Marzano check - circle: 1 2 3